

TravelEase Hub: An AI-Powered Travel Assistance Platform with Pooling and Smart Expense Splitting

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Abstract—TravelEase Hub is proposed as a full-stack, AI-driven travel assistance platform that addresses shortcomings in contemporary travel applications by combining personalization, collaborative trip planning, and automated financial reconciliation for group travel. The platform integrates AI itinerary generation, budget estimation, user pooling (matching travelers with similar routes and schedules), and automated expense splitting into a single secure system. This paper presents the system design, data and model choices, evaluation methodology, and experimental results from a usability study and performance benchmarks. The work emphasizes privacy-preserving design choices, extensibility, and pragmatic trade-offs for real-world deployment.

Index Terms—TravelTech, Recommender Systems, Group Travel, Expense Splitting, AI Itinerary Planning, Supabase, TensorFlow.js

I. INTRODUCTION

Digital travel services have revolutionized how travelers discover, plan, and purchase travel-related products. Leading sites and apps optimize search, present curated bundles, and enable bookings; however, most remain oriented toward transactional flows rather than adaptive, collaborative planning. Users tasked with organizing group travel frequently encounter three recurring problems: (1) aligning schedules and itineraries across multiple participants, (2) fairly splitting and tracking shared expenses, and (3) obtaining personalized itineraries that adapt to preferences, constraints, and real-time context.

TravelEase Hub targets these gaps by bringing multi-modal AI capabilities into the travel workflow: personalized itinerary recommendation, pooled-user discovery (for matching travelers who can share resources), and smart expense splitting that produces transparent, auditable cost allocations. The platform also offers travel agencies administrative capabilities and supports verification workflows for published packages. The system is implemented as a three-tier architecture with a React-based presentation layer, a Node.js/Express application layer, and a Supabase/PostgreSQL data layer, while AI components use lightweight models (TensorFlow.js) and cloud APIs where needed. Security and privacy concerns are addressed via JWT-based authentication, role-based access control, and recommendations for encryption-at-rest in production deployments.

This paper expands on design decisions, related work, evaluation methodology, and empirical findings from prototype

testing with real users and simulated workloads.

II. LITERATURE REVIEW

A broad literature exists across recommender systems, group decision-making, digital tourism, and expense-management. Recommender system surveys lay out collaborative, content-based, and hybrid approaches that underpin modern personalization engines [1], [2]. In travel-specific contexts, itinerary recommendation and POI (point-of-interest) suggestion literature has advanced using matrix factorization, sequential modeling, and deep learning to capture temporal and contextual preferences [3]–[5].

Group travel and collaborative decision-making have been explored in the context of social choice and multi-stakeholder recommendation; solutions often aim to reconcile heterogeneous preferences using aggregation strategies or fair-share optimization [6], [7]. Pooling and ride-sharing literature provides transferable methods for matching users based on spatio-temporal similarity and cost-optimization [8], [9].

Expense splitting has been addressed by several commercial and academic works that propose algorithms for minimal-transaction settlement, fairness criteria, and privacy-preserving ledgering [10], [11]. Research on blockchain-backed verification in travel has proposed immutable receipts and tamper-evident vaccination or ticket records, though practical adoption remains limited [12], [13].

Tourism research highlights the trend toward smart tourism ecosystems, integrating IoT, AI, and context-aware services to create seamless user experiences [14]. Personalization is further enhanced by using predictive analytics and social recommendation methods [15]–[17]. Despite these advances, current commercial applications rarely combine itinerary personalization with collaborative features such as pooling and automatic expense reconciliation.

This paper synthesizes these lines of work into a single integrated platform for individual and group travel. Compared to existing travel apps (MakeMyTrip, Expedia, Booking.com), TravelEase Hub emphasizes collaborative planning and automated financial reconciliation, bringing research ideas into a cohesive, user-centered application.

III. SYSTEM DESIGN AND METHODOLOGY

We organize the design discussion around architecture, data handling, AI modules, and security/privacy.

A. Three-Tier Architecture

The system follows a standard three-tier design:

- 1) **Presentation Layer:** A cross-platform frontend implemented in React (Expo/React Native where mobile preview is required) provides dashboards for three roles (traveler, agency, admin). Key UX elements include package search with smart filters, an AI itinerary composer, pooling toggles, and an expense settlement console.
- 2) **Application Layer:** Node.js with Express orchestrates business logic: booking lifecycle, pooling matching, settlement calculations, notifications, and AI inference orchestration. Stateless microservices are favored to scale workload; ephemeral caches reduce repeated computation for common queries.
- 3) **Data Layer:** Supabase (PostgreSQL) persists user profiles, package metadata, booking events, and ledger entries for expenses. Role-based access control (RBAC) and JWT tokens authenticate and authorize requests.

B. Data Collection and Preprocessing

Primary data sources are user-provided: travel preferences, past booking history, ratings, and explicit constraints (budget, dates). Secondary datasets include destination metadata (POIs, opening hours), transportation schedules, hotel availability, and public holiday calendars. Preprocessing steps include normalization, embedding construction, and anomaly detection.

C. AI and Algorithms

Key algorithmic modules:

- **Itinerary Recommendation:** A hybrid approach combining content-based filters, collaborative signals, and sequence-aware models to propose day-wise activities within budget and duration constraints.
- **Pooling Matcher:** A clustering-based matcher computing spatio-temporal similarity scores and preference overlap.
- **Expense Splitting:** An optimization routine minimizing settlement transactions while ensuring fairness.
- **Ranking and Personalization:** A gradient-boosted trees + neural ranking ensemble for accuracy and explainability.

D. Security and Privacy

Security considerations include JWT tokens, RBAC, encryption-at-rest, opt-in pooling, and GDPR-style data management.

IV. EVALUATION

The platform was evaluated through functional testing, performance benchmarking, and user feedback collection.

A. Usability Study

A pilot study with 36 participants across 9 groups was conducted. Metrics measured were task completion time, satisfaction scores, and adoption intent. Results showed itinerary generation reduced planning time by over 40%, while expense automation improved transparency.

B. Performance Benchmarking

Load testing showed API latencies below 600ms for itinerary generation, with pooling match computation remaining sub-second at moderate scale.

V. RESULTS

Key findings include:

- 82% of users rated itineraries highly relevant.
- Pooling was accepted by 67% of users, with majority confirming utility.
- Settlement optimization reduced required transactions by 41%.

VI. CONCLUSION AND FUTURE WORK

TravelEase Hub demonstrates how AI-driven personalization, pooling, and automated expense splitting can materially improve group travel planning. Future work includes deeper integration with predictive pricing, differential privacy for analytics, blockchain-backed settlements, and real-world pilot deployments with travel agencies.

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REFERENCES

- [1] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions," *IEEE TKDE*, vol. 17, no. 6, pp. 734–749, 2005.
- [2] F. Ricci, L. Rokach, B. Shapira, and P. B. Kantor, *Recommender Systems Handbook*. Springer, 2011.
- [3] S. Zhang, L. Yao, A. Sun, and Y. Tay, "Deep learning based recommender system: A survey and new perspectives," *ACM Comput. Surv.*, vol. 52, no. 1, pp. 1–38, 2019.
- [4] D. Wang, H. Wang, K. Zhao, and L. Pan, "Next location prediction with spatial-temporal attentions," in *Proc. ACM SIGSPATIAL*, 2018, pp. 1–10.
- [5] X. Liu, J. Zhao, and K. Zheng, "Itinerary planning via sequence modeling for POI recommendation," *Inf. Sci.*, vol. 512, pp. 1234–1250, 2020.
- [6] J. Masthoff, "Group recommender systems: Survey and challenges," in *Recommender Systems Handbook*, Springer, 2011, pp. 717–741.
- [7] R. Burke, "Hybrid recommender systems: Survey and experiments," *User Model. User-Adapted Interact.*, vol. 12, no. 4, pp. 331–370, 2002.
- [8] J. Alonso-Mora, et al., "On-demand high-capacity ride-sharing via dynamic trip-vehicle assignment," *PNAS*, vol. 114, no. 3, pp. 462–467, 2017.
- [9] Y. Ge, Y. Zheng, and S. Qi, "Ride-sharing matching with spatio-temporal constraints: A survey," *ACM Comput. Surv.*, vol. 52, no. 6, 2020.
- [10] J. Karp and M. T. Nguyen, "Automating group expense settlements: algorithms and fairness," in *Proc. Intl. Conf. Financial Informatics*, 2018, pp. 45–54.
- [11] I. Shah and K. Narayan, "Minimizing transactions for group expense settlement," *J. Syst. Softw.*, vol. 130, pp. 128–138, 2017.

- [12] G. Wood and R. Kumar, "Blockchain for travel: use-cases and challenges," *Tourism Management*, vol. 70, pp. 1–10, 2019.
- [13] Z. Zheng, S. Xie, and H. Dai, "Blockchain challenges and opportunities: A survey," *Int. J. Web Grid Serv.*, vol. 14, no. 4, pp. 352–375, 2018.
- [14] U. Gretzel, J. Sigala, B. Xiang, and E. Koo, "Smart tourism: foundations and developments," *Electron. Markets*, vol. 25, pp. 179–188, 2015.
- [15] S. Mehta, "Advances in artificial intelligence for personalization in tourism," *Tourism Inf. Technol. J.*, vol. 12, no. 1, pp. 23–42, 2025.
- [16] I. Papastratis, "AI recommendation engines for travel personalization," *Scientific Reports*, vol. 14, 2024.
- [17] W. H. Kwok, "Group travel and AI: fairness and optimization," *Behav. Res. Technol.*, vol. 8, no. 2, pp. 99–115, 2024.